NATURAL LANGUAGE PROCESSING

Assignment

Completed by:

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Section: D

01

WORD EMBEDDING SIMILARITY

How can machines mathematically compute similarities between concepts, identify categories, and solve word analogies?

How can we represent words numerically and measure how similar two words are?

Word Embeddings (words → vectors):

Operate on distributional hypothesis: word used in similar contexts tend to have similar meanings.

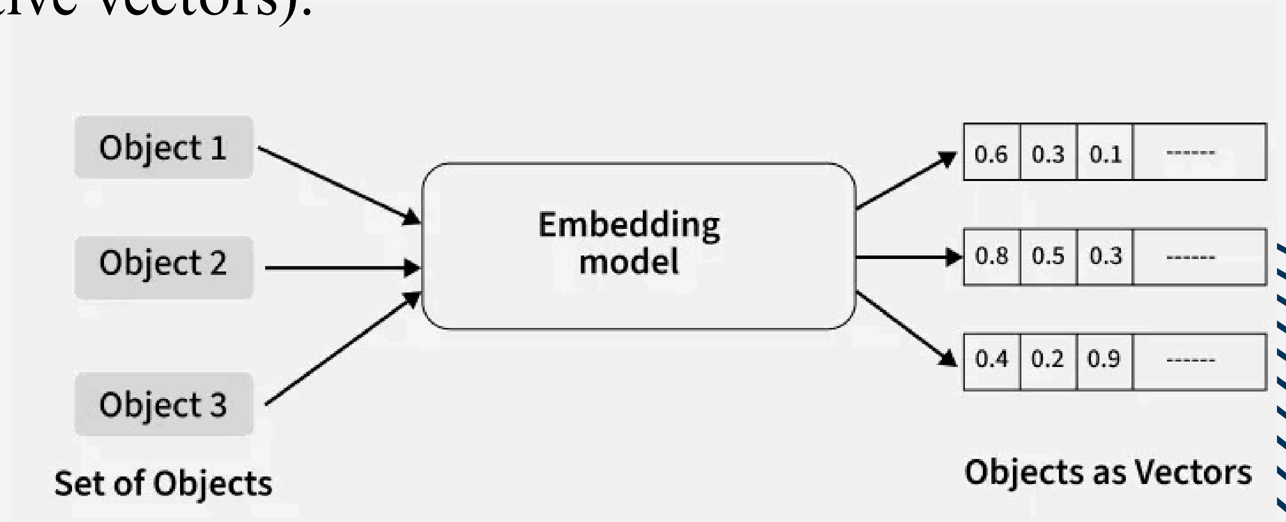
Word2Vec model (pre-trained model):

Maps each word into a dense and continuous vector space (here, a 300 dimensional space).

Cosine Similarity:

To determine, how similar two words (v1 and v2) are (measure angle between their respective vectors):

Cosine Similarity = v1 . v2 / |v1| |v2|



Dimensionality Reduction (PCA):

To plot word relationship on 2D scatter plot, 300 dimensions reduced to 2D using PCA.

WORKFLOW

Step 1: Load pretrained model.

Step 2: Similarity function

Step 3: Testing basic word pairs

Step 4: Category-based Analysis

Step 5: Vecctor Arithmetic (semantic search and analogies)

Step 6: Dimensionality Reduction and 2D plotting

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KEY FINDINGS AND ANALYSIS

Basic Similarity Testing

```
king       - queen     : 0.6511
doctor     - nurse     : 0.6320
car        - tree      : 0.2711
man        - woman     : 0.7664
paris      - france    : 0.5551
apple      - fruit     : 0.6410
dog        - cat       : 0.7609
computer   - keyboard  : 0.3964
sun        - moon      : 0.4263
book       - car       : 0.1279
```

```
Top Similar Pairs:
man-woman : 0.7664
dog-cat : 0.7609
king-queen : 0.6511
apple-fruit : 0.641
doctor-nurse : 0.632

Least Similar Pairs:
paris-france : 0.5551
sun-moon : 0.4263
computer-keyboard : 0.3964
car-tree : 0.2711
book-car : 0.1279
```

Score:
High → related
low → unrelated

```
Similar words to 'king':

kings           0.7138
queen           0.6511
monarch         0.6413
crown_prince    0.6204
prince          0.6160
```

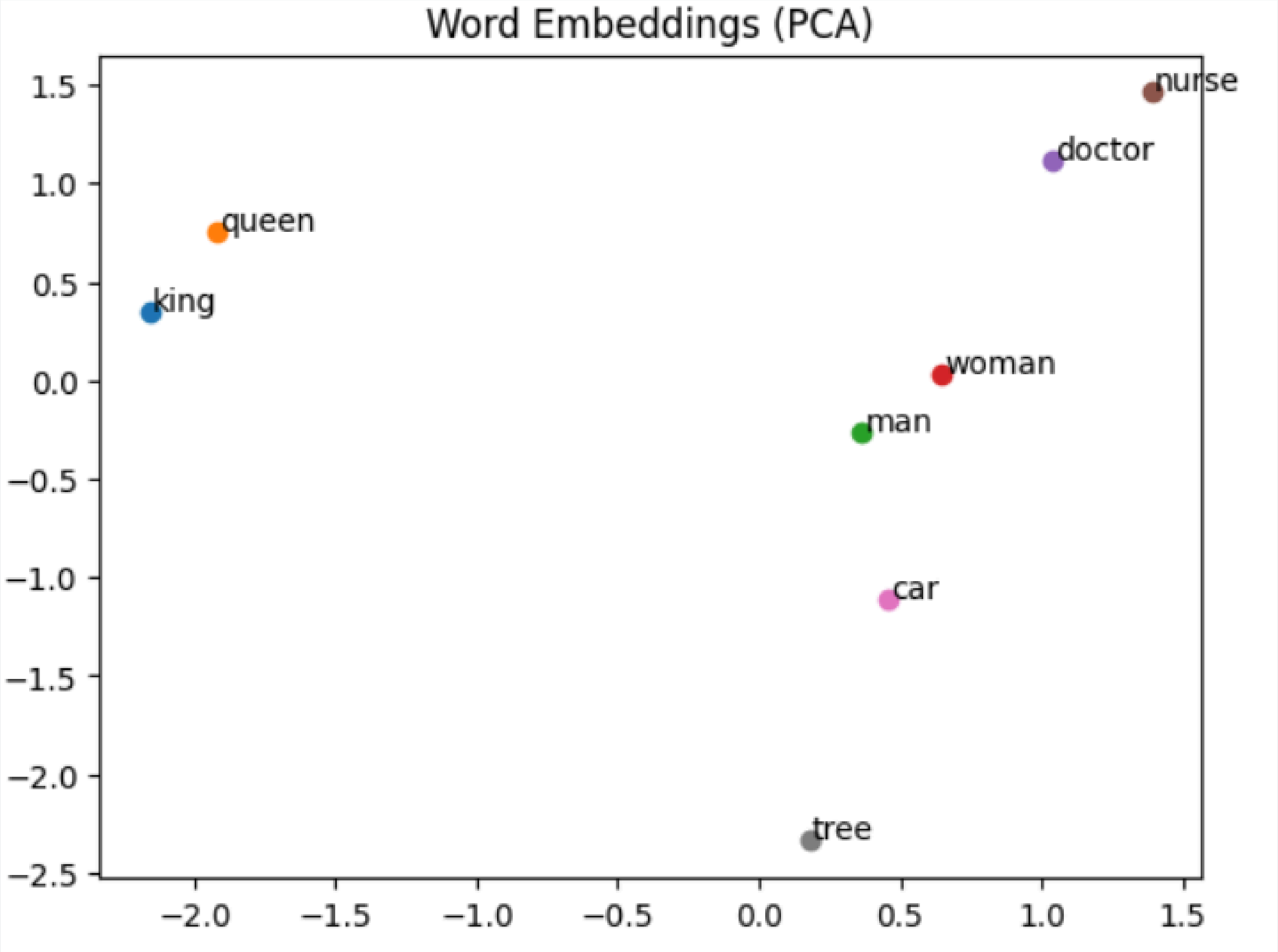
Word Analogies (Vector Arithmetic)

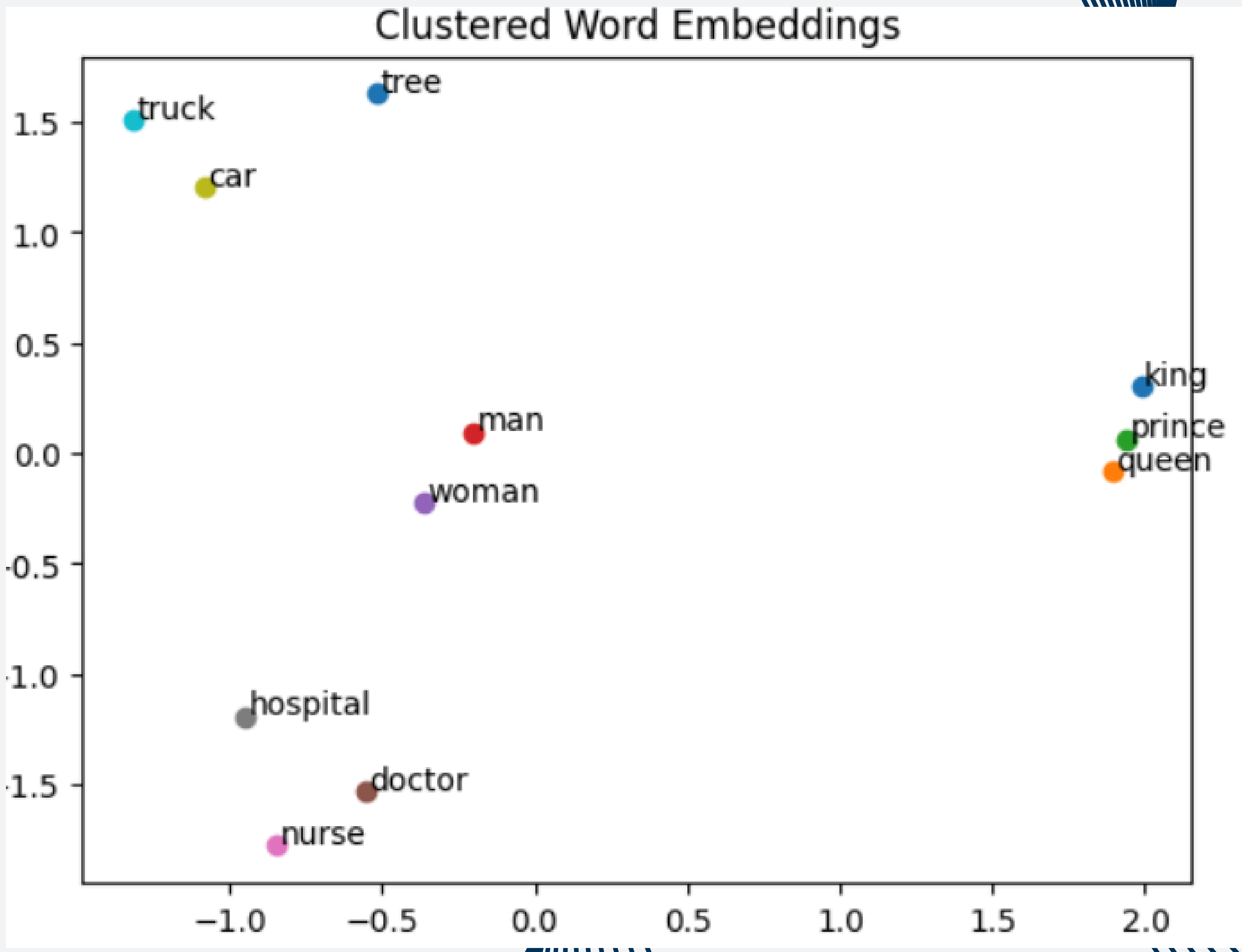
```
['king', 'woman'] - ['man'] =>
queen           0.7118
monarch         0.6190
princess        0.5902

['italy', 'paris'] - ['france'] =>
lohan           0.5070
madrid          0.4818
heidi           0.4800

['boy', 'girl'] - ['man'] =>
toddler         0.6190
daughter        0.5895
child           0.5891

['walking', 'ran'] - ['walk'] =>
running         0.5351
went            0.5181
sped            0.5003
```

```
king - man + woman ≈

queen           0.7118
monarch         0.6190
princess        0.5902
crown_prince    0.5499
prince          0.5377
```

Relationships encoded as vectors.

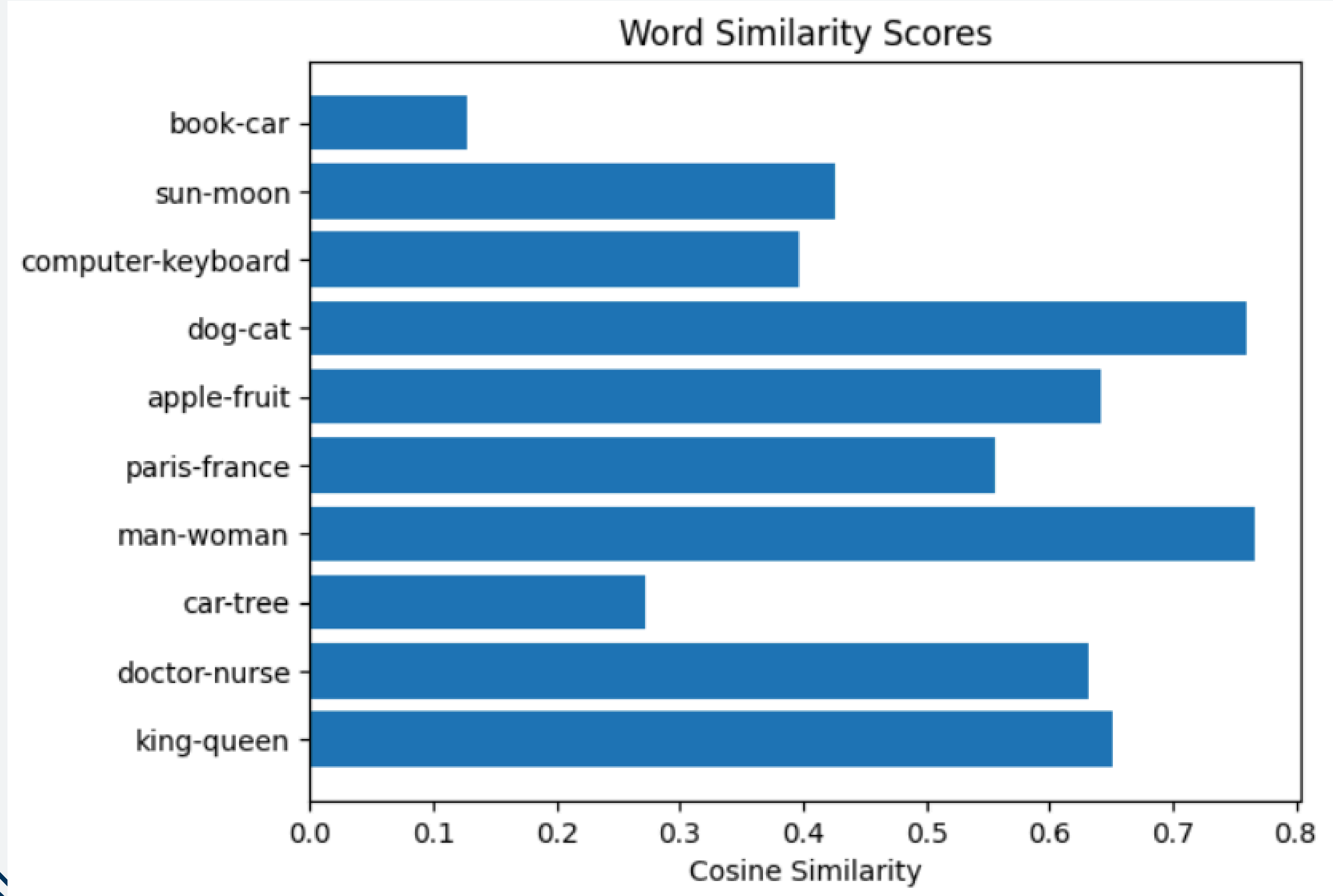
VISUALIZATIONS

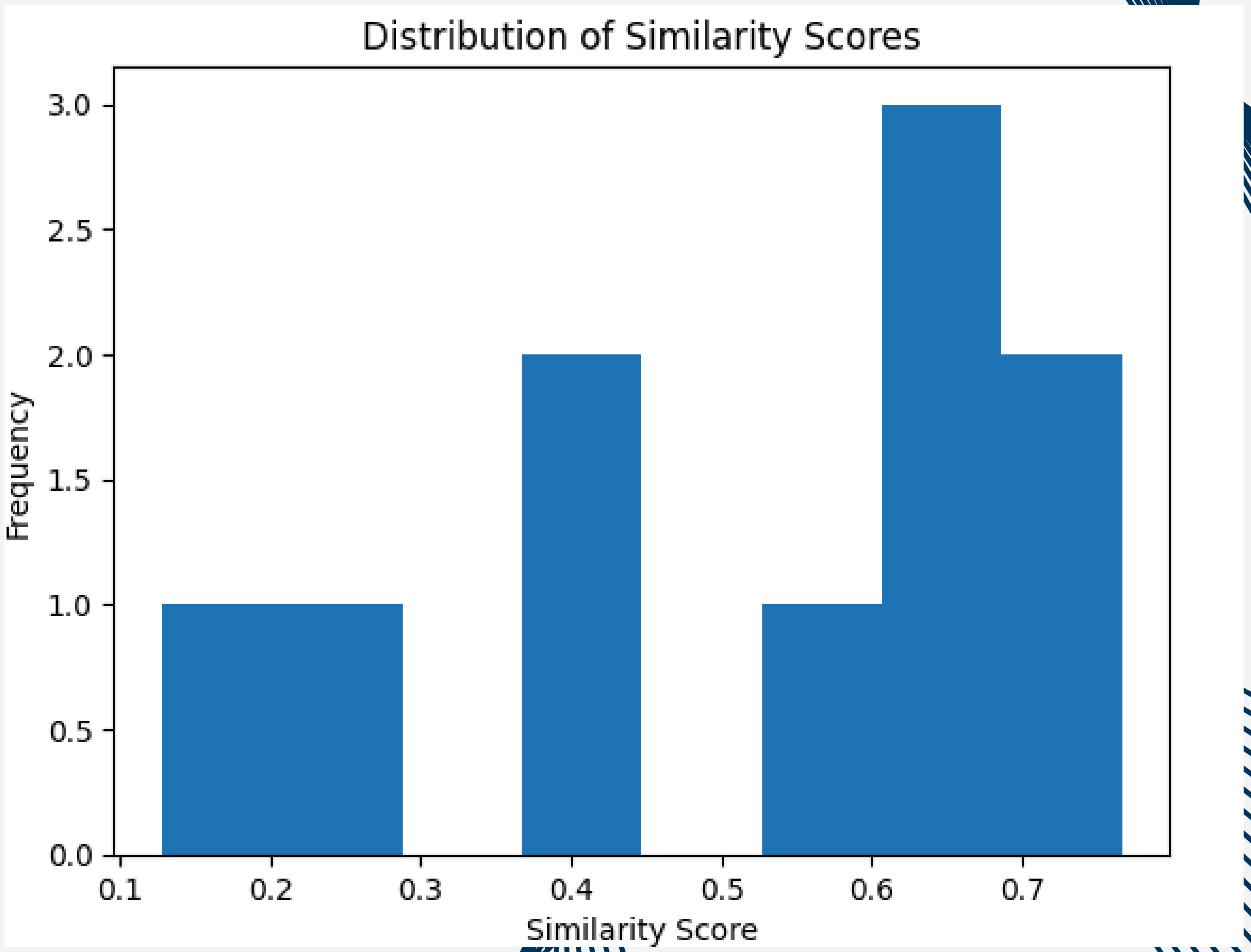




Similar words cluster together.
-eg. king, queen, prince are close
car, tree far apart

DISTRIBUTION PLOTS





- Most similarities between 0.3- 0.7
- i.e. Most words share some context
- Completely unrelated words are rare.

TEST CASES

```
Enter first word (or 'exit'): machine
Enter second word: gear
Similarity: 0.2778
Enter first word (or 'exit'): screw
Enter second word: screwdriver
Similarity: 0.4315
Enter first word (or 'exit'): water
Enter second word: milk
Similarity: 0.4049
Enter first word (or 'exit'): milk
Enter second word: curd
Similarity: 0.4895
Enter first word (or 'exit'): usa
Enter second word: washington
Similarity: 0.56
Enter first word (or 'exit'): pen
Enter second word: pen
Similarity: 1.0
Enter first word (or 'exit'): paint
Enter second word: brush
Similarity: 0.3416
Enter first word (or 'exit'): pepsi
Enter second word: coke
Similarity: 0.4256
Enter first word (or 'exit'): 1
Enter second word: 3
Similarity: 0.8894
Enter first word (or 'exit'): 10
Enter second word: 100
```

Model captures multiple relationship types.

Interesting observations:

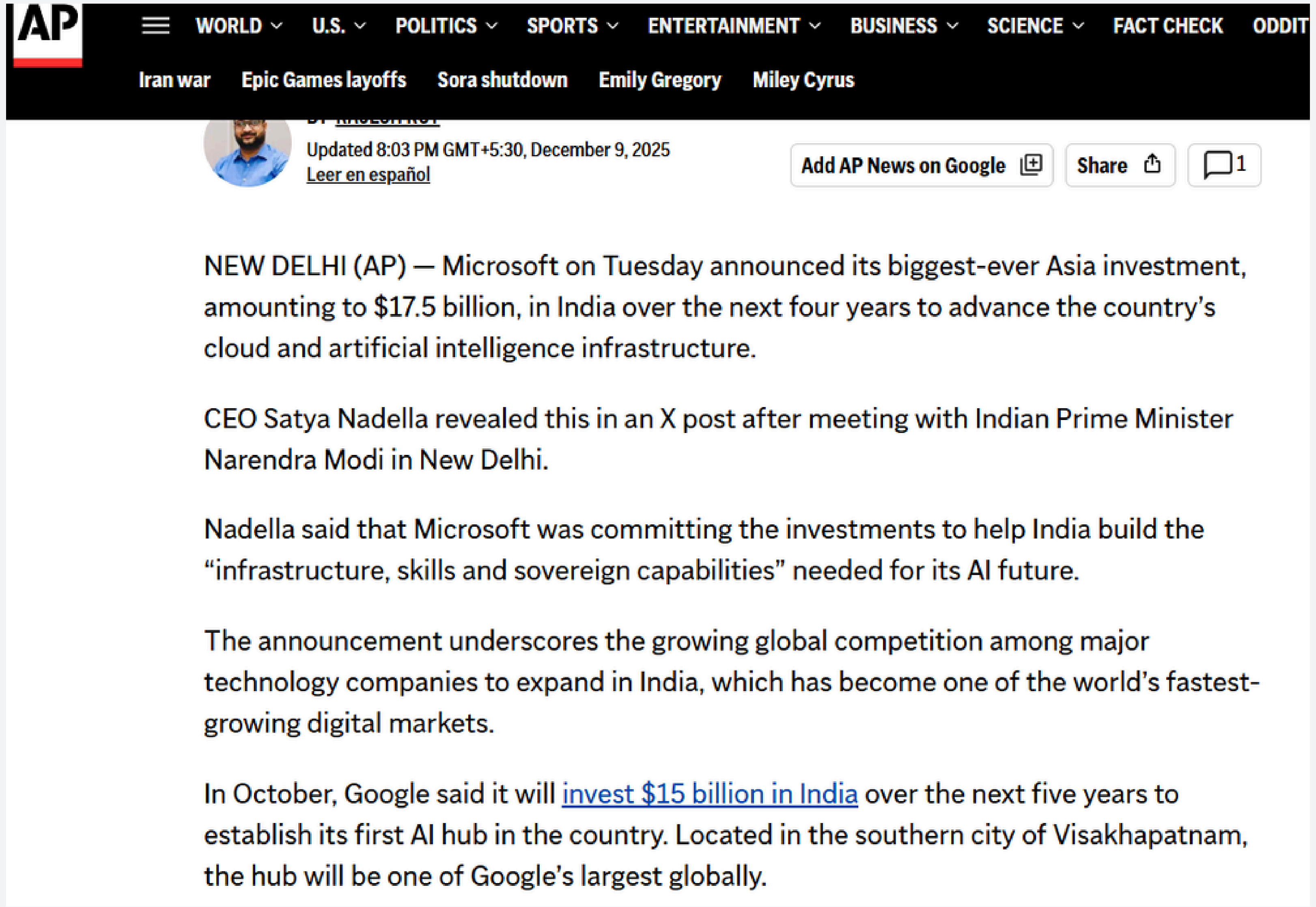
- Numbers appear in similar context (1-3 → 0.8894)
- Words2Vec treats words individually; cannot understand phrases (machine-learning → 0.058)

Challenges

- Vocabulary limitations
- Case sensitivity
- No phrase understanding
- Context-based, not dictionary based

02

NAMED ENTITY RECOGNITION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NEW DELHI (AP) — Microsoft on Tuesday announced its biggest-ever Asia investment, amounting to $17.5 billion, in India over the next four years to advance the country's cloud and artificial intelligence infrastructure.

CEO Satya Nadella revealed this in an X post after meeting with Indian Prime Minister Narendra Modi in New Delhi.

Nadella said that Microsoft was committing the investments to help India build the "infrastructure, skills and sovereign capabilities" needed for its AI future.

The announcement underscores the growing global competition among major technology companies to expand in India, which has become one of the world's fastest-growing digital markets.

In October, Google said it will invest $15 billion in India over the next five years to establish its first AI hub in the country. Located in the southern city of Visakhapatnam, the hub will be one of Google's largest globally.

A brief overview of the task: Extracting Persons, Organizations, and Locations from a Microsoft/India news article using NLTK Library

Step 1: Tokenization
Step 2: POS Tagging (Required before NER)
Step 3: Named Entity Chunking

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BASE RESULTS

```
--- NER Analysis Results ---
Counted Persons: 8 -> ['Microsoft', 'Narendra Modi', 'Nadella', 'Microsoft', 'Google', 'Google', 'Nadella', 'Bengaluru']
Counted Organizations: 4 -> ['NEW', 'DELHI', 'AP', 'CEO Satya Nadella']
Counted Locations: 10 -> ['Asia', 'India', 'Indian', 'New Delhi', 'India', 'India', 'India', 'Visakhapatnam', 'India', 'Mumbai']
```

4. Visualizing the Tree

```
chunk_tree.pretty_print()
```

```
...
GPE      PERSON               GPE          PERSON     PERSON      GPE       GPE       PERSON      GPE        GPE          PERSON      PERSON     GPE        PERSON         GPE
 |      ________|______       ____|_____      |          |          |         |          |          |          |             |           |         |            |              |
ian/JJ Narendra/NNP    Modi/NNP New/NNP  Delhi/NNP Nadella/NNP Microsoft/NNP India/NNP India/NNP Google/NNP India/NNP Visakhapatnam/ Google/NNP Nadella/NN India/NNP Bengaluru/NNP Mumbai/NNP
                                                                                                              NNP
```

Note: NLTK was designed for teaching and research, not for high-accuracy production apps. Its NER module hasn't been significantly updated in years, which is why it thinks "Google" is a Person

APPLIED REGEX RULES

Regular Expressions are powerful text patterns used to find, match, and manipulate specific sequences of characters within larger texts. In our task, we defined a 'grammar' using regex to instruct NLTK on how to recognize patterns for Persons, Organizations, and Titles based on their part-of-speech tags

```python
# 1. Automated Grammar: Just find sequences of Proper Nouns (NNP)
grammar = r"""
  ENTITY: {<NNP>+}   # Grab EVERY sequence of capitalized words as one unit
"""

def extract_entities_final_version(tags, grammar):
    parser = RegexpParser(grammar)
    tree = parser.parse(tags)

    entities = {"ORGANIZATION": [], "PERSON": [], "LOCATION": []}

    # Automated Logic based on standard English patterns
    for subtree in tree:
        if isinstance(subtree, Tree):
            # 1. Join the words
            full_name = " ".join([leaf[0] for leaf in subtree.leaves()])

            # 2. CLEANING: Remove punctuation like '’' or '(AP)'
            full_name = re.sub(r"[^a-zA-Z\s]", "", full_name).strip()
            if not full_name or full_name in ['Tuesday', 'October', 'AP', 'X']:
                continue

            # 3. SMART SORTING:
            # - If it's 2-3 words and not 'New Delhi', it's almost always a PERSON
            if len(full_name.split()) >= 2:
                if "Delhi" in full_name:
                    entities["LOCATION"].append(full_name)
                else:
                    entities["PERSON"].append(full_name)

            # - If it's a single word:
            else:
                # Common sense check for Locations
                if full_name in ['India', 'Mumbai', 'Bengaluru', 'Visakhapatnam', 'Asia', 'DEHLI']:
                    entities["LOCATION"].append(full_name)
                # If it's 'AI', it's a technology, usually categorized as ORG or skipped
                elif full_name == 'AI':
                    continue
                elif full_name == 'DELHI':
                    continue
                else:
                    entities["ORGANIZATION"].append(full_name)

    # Unique values only
    for key in entities:
        entities[key] = sorted(list(set(entities[key])))

    return entities
```

Results:

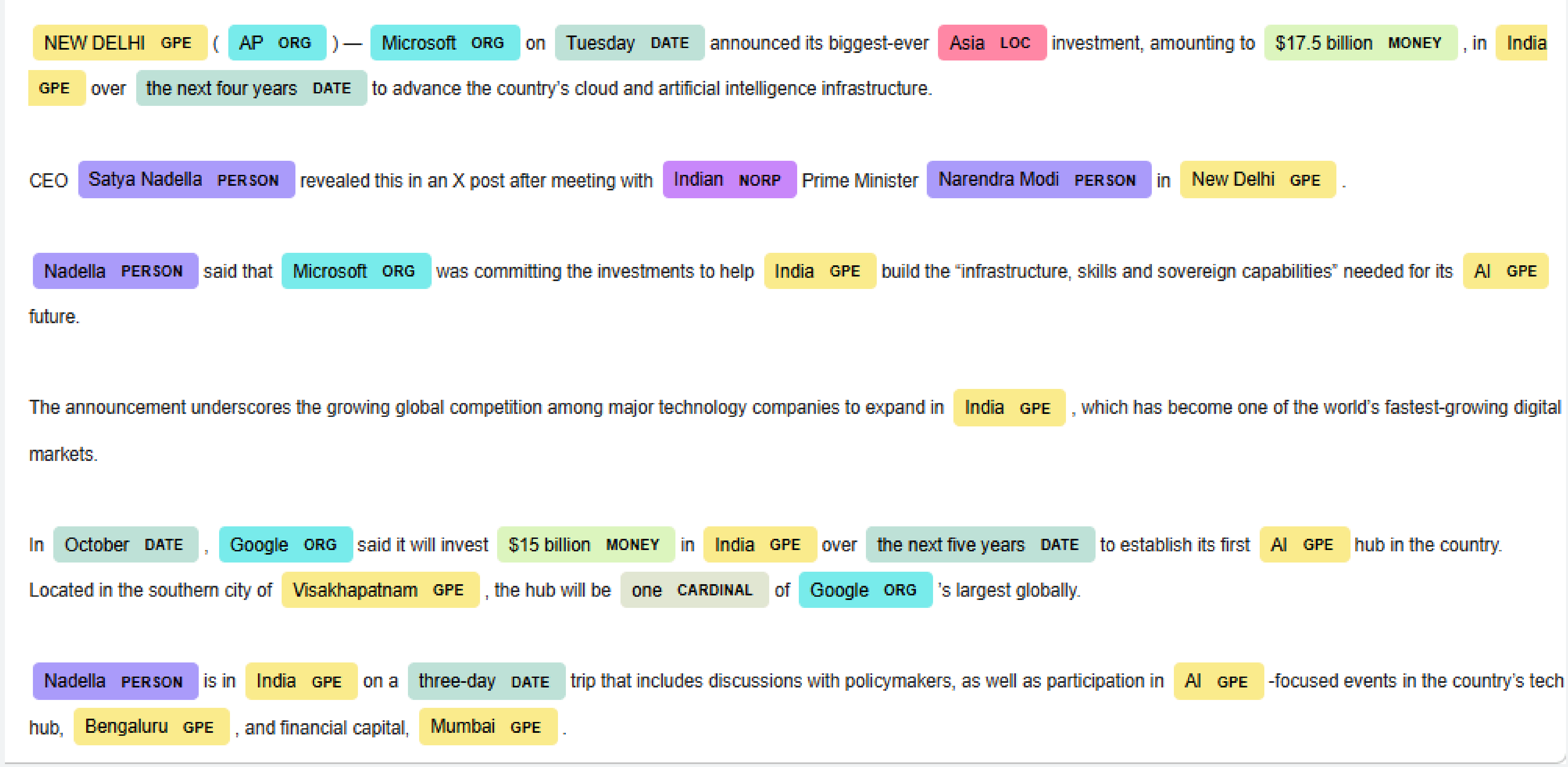
```
Organizations: ['Google', 'Microsoft', 'Nadella']
Persons:       ['CEO Satya Nadella', 'Prime Minister Narendra Modi']
Locations:     ['Asia', 'Bengaluru', 'India', 'Mumbai', 'New Delhi', 'Visakhapatnam']
```

Tree Visualisation:

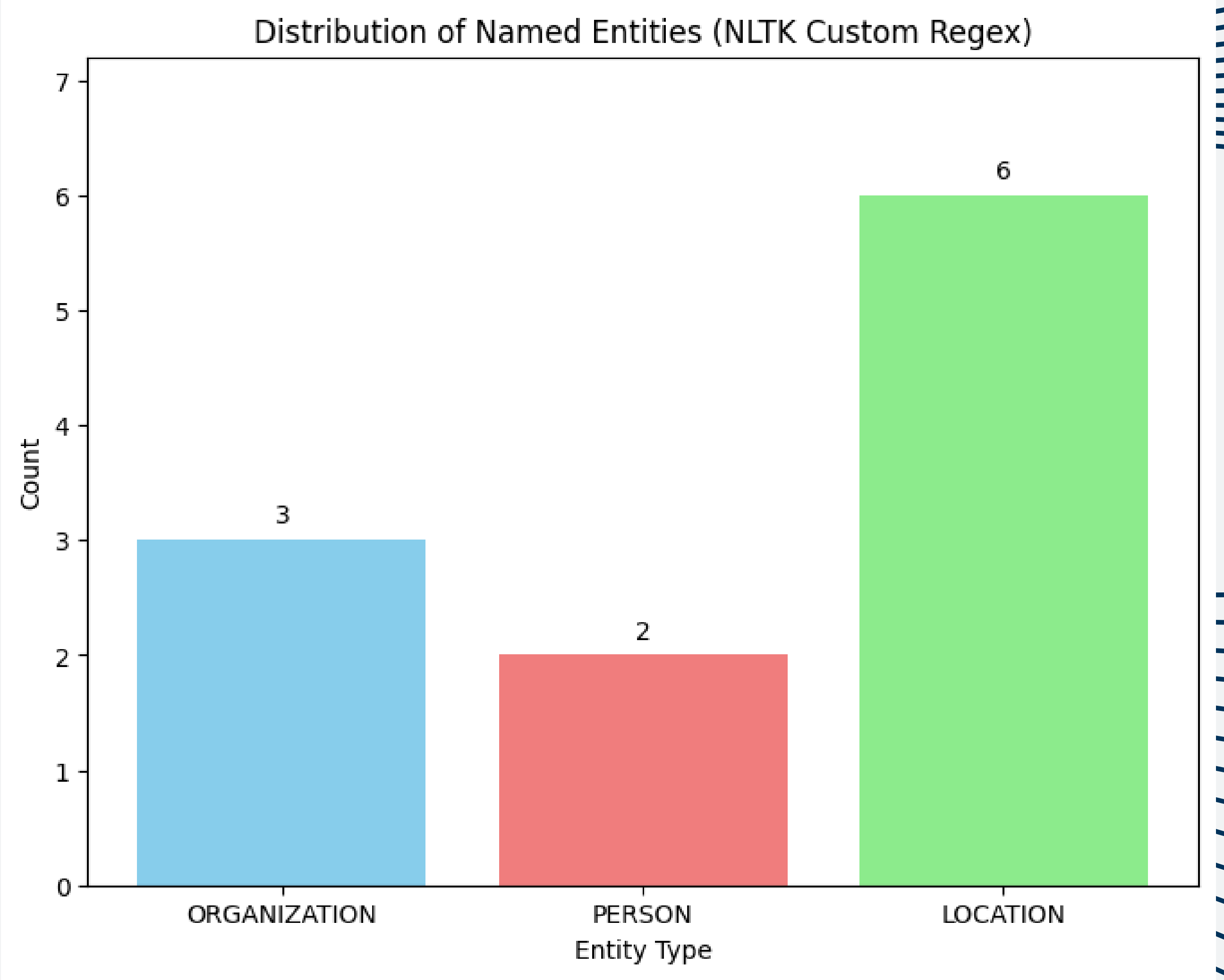
```
tree.pretty_print()

                                    ENTITY          ENTITY      ENTITY      ENTITY  ENTITY   ENTITY     ENTITY     ENTITY     ENTITY  ENTITY    ENTITY                 ENTITY         ENTITY  ENTITY
_______________           ______|_______            |           |           |       |        |          |          |          |       |         |               ______|_____          |       |
Narendra/NNP Modi/NNP New/NNP       Delhi/NNP Nadella/NNP Microsoft/NNP India/NNP AI/NNP India/NNP October/NNP Google/NNP India/NNP AI/NNP Visakhapatnam/ Google/NNP        ’/NNP India/NNP ’/NNP  B
                                                                                                                                  NNP
```

VISUALISATION



Using DisplaCy SpaCy Builtin Tool

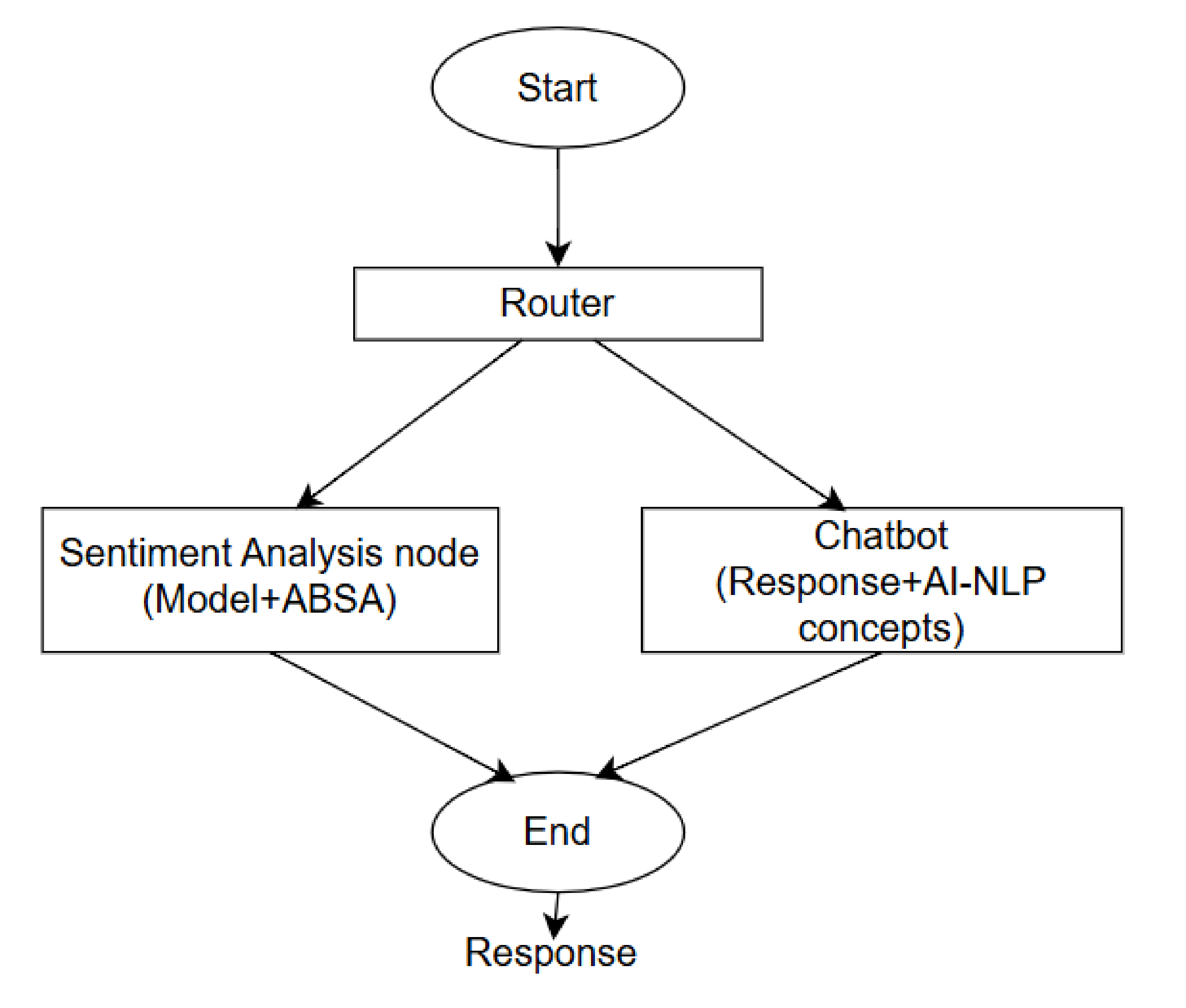


Distribution Bar Graph

HYBRID CHATBOT AND ASPECT-BASED SENTIMENT ANALYSIS ARCHITECTURE

Conversational AI and Opinion Mining for Product Reviews

- A hybrid architecture that acts as single entry point for user text.
- It routes queries to chatbot and ABSA module as required, and handles both conversational intents and complex sentiment reviews.



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Let's talk about each module in detail.

- Sentiment Analysis & Aspect-Based Sentiment Analysis (ABSA)- **Anwesha Singh**
- Chatbot on NLP concepts and AI applications - **Oshika Sharma**

SECTION 1: SENTIMENT ANALYSIS & ASPECT-BASED SENTIMENT ANALYSIS (ABSA)

- Part of Hybrid Chatbot System
- NLP-Based Opinion Mining
- Focus: E-commerce Product Reviews

Problem:

Fine grained opinions.

Mixed sentiments in same sentence.

Single label for entire text, ignores multiple opinions, and feature-specific feedback.

Need **Aspect-Based Sentiment Analysis (ABSA)**

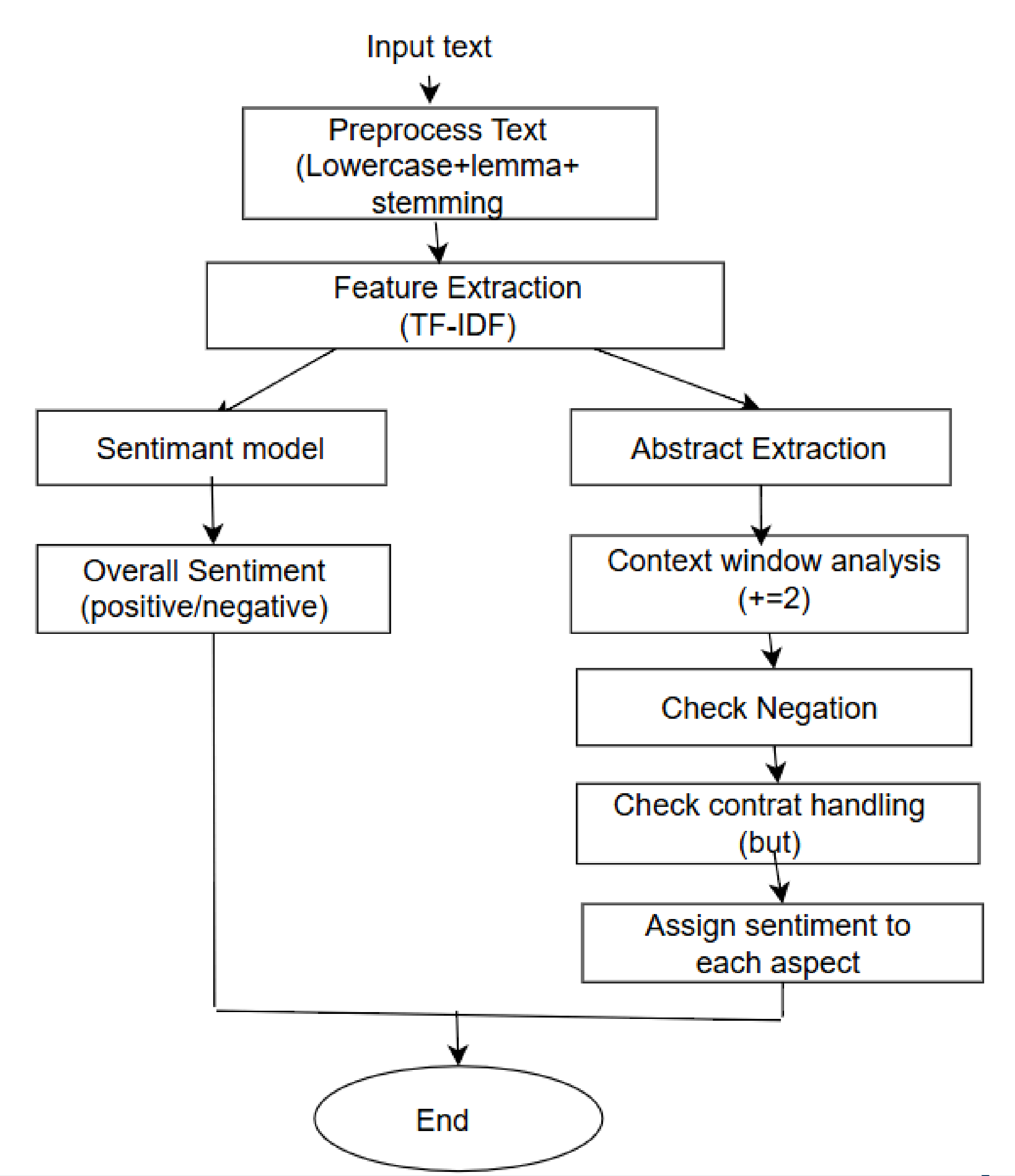
Concepts to know:

Sentiment Analysis: Technique to determine polarity/emotion of the text.

Types: Document-level, Sentence-level, **Aspect-level (ABSA)**

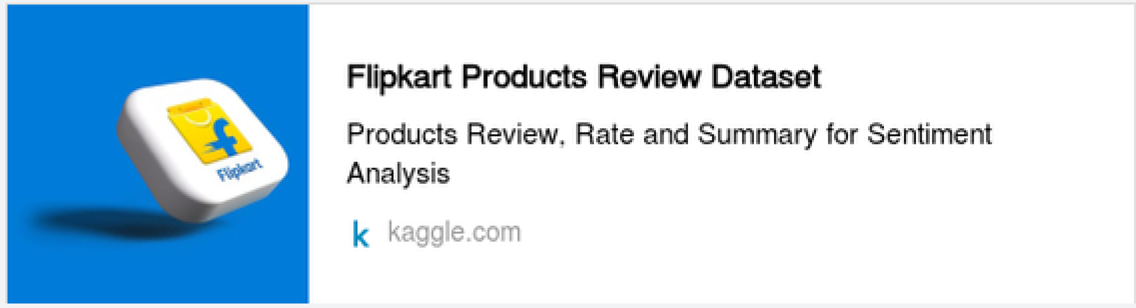
ABSA: Sentiment for each aspect, more fine grained and accurate analysis. Works well for E-commerce review analysis, and Customer feedback monitoring

Input → Router



DATASET

Flipkart Product Review Dataset



I. PREPROCESSING

1. Lowercasing
2. Stopword Removal
3. Puctuation Removal
4. Lemmatization

Reduces noise and dimensionality

II. FEATURE EXTRACTION (TF-IDF)

- Text → Numerical Vectors
- TF : word frequency in document
- IDF : rarity of word across corpus (importance to rare nut meaningfull words
- TF-IDF = TF × IDF

III. SENTIMENT CLASSIFICATION MODEL

Model used: **Multinomial Naive Bayes**

Output: Positive/ Negative

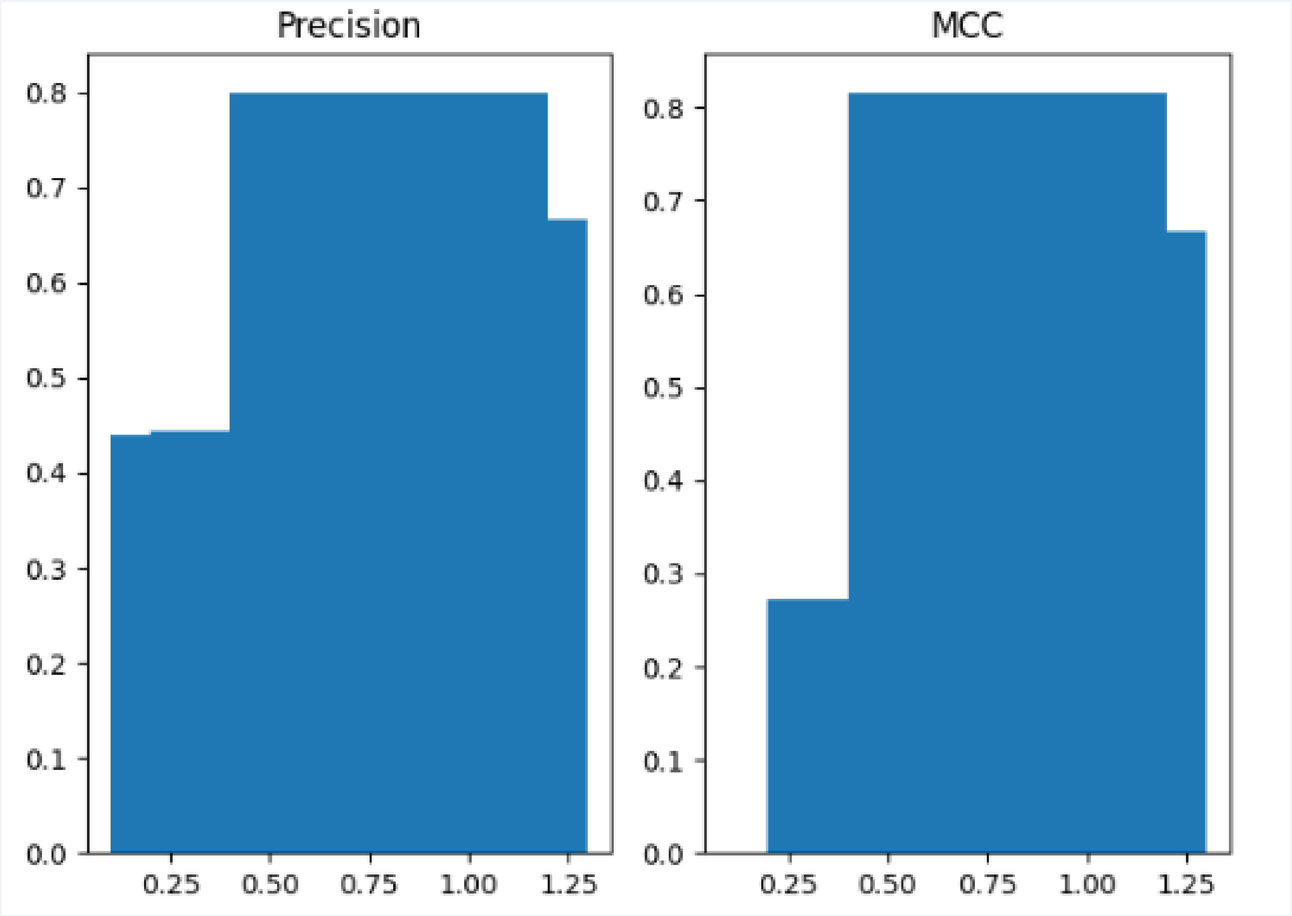
IV. ABSA MODEL

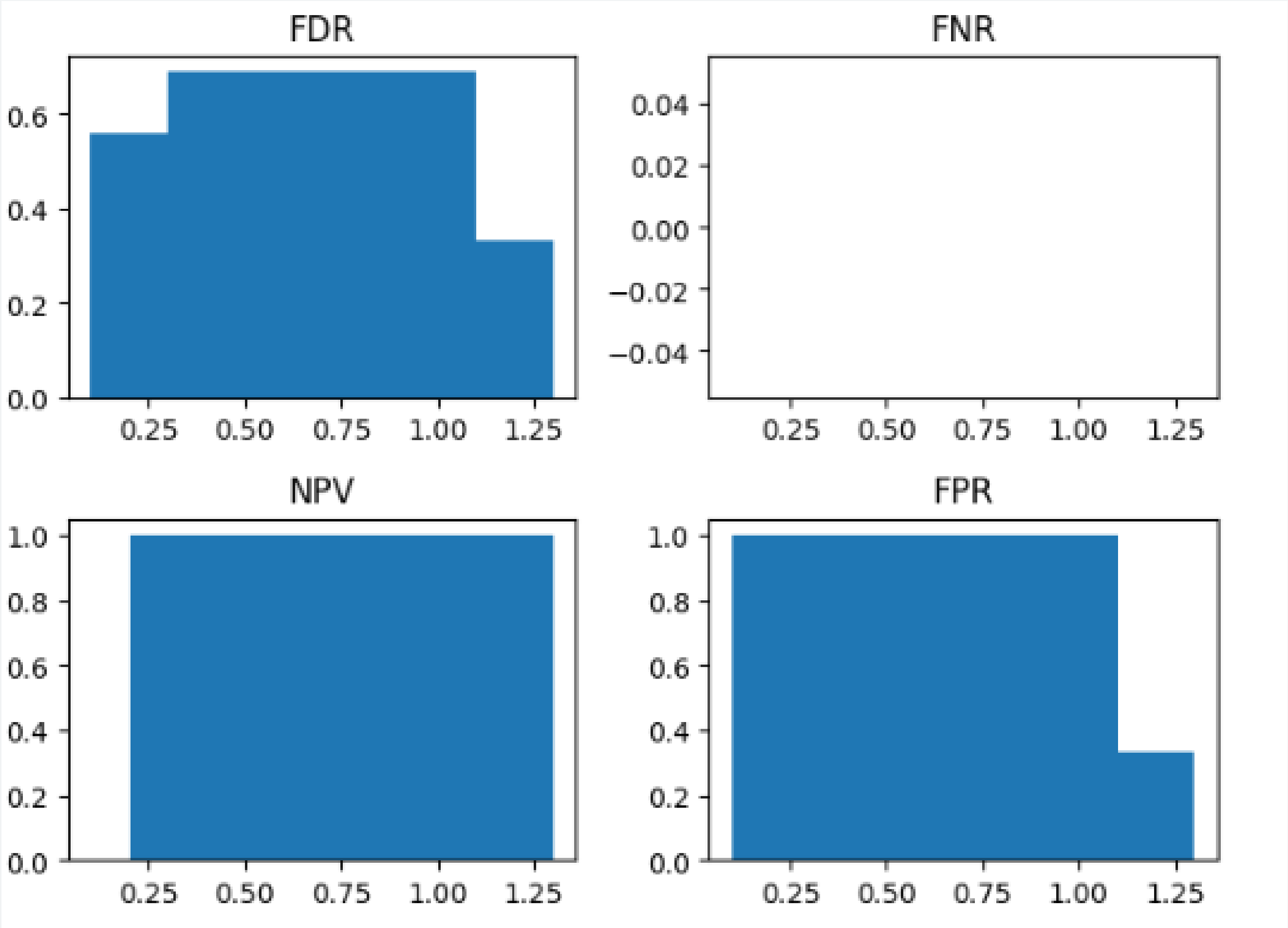
- Custom rule-based system
- Instead of heavy Deep learning models, as they cause hallucinations and overfitting with small datasets

OUTPUTS WITHOUT ABSA

```
Confusion Matrix:
 [[5 1]
 [0 4]]
Accuracy: 0.9
Precision: 0.8
Recall: 1.0
Specificity: 0.8333333333333334
F1 Score: 0.8888888888888888
FPR: 0.16666666666666666
FNR: 0.0
NPV: 1.0
FDR: 0.2
MCC: 0.816496580927726
```

Using Multinomial Naive-Bayes





ASPECT-BASED SENTIMENT ANALYSIS (ABSA)

1. Identify product features

```
def extract_aspects(text):
    words = text.split()
    return [w for w in words if w not in positive_words and w not in negative_words]
```

2. Context Window Analysis

- Window size = ±2 words
- Takes context of previous and upcoming words, to ensure correct sentiment analysis.

3. Negation and Contrast handling

- Sentiment flip due to: not, don't, didn't
- Detect word: 'but'; prevents sentiment mixing.
  eg. Cooling good but noise high.

4. Final ABSA output

   Produces Aspect-wise sentiment

```
Review: mind blowing purchase best quality beat price thank flipkart
Aspects: {'blowing': 'Positive', 'purchase': 'Positive', 'quality': 'Positive', 'beat': 'Positive'}
Review: nan size perfect
Aspects: {'nan': 'Positive', 'size': 'Positive'}
Review: poor bad product working properly
Aspects: {'product': 'Negative', 'working': 'Negative'}
Review: terrible product good
Aspects: {}
Review: super smooth delivery installation amazing product produce less noise
Aspects: {'delivery': 'Positive', 'installation': 'Positive', 'product': 'Positive', 'produce': 'Positive'}
Review: nan bad product
Aspects: {'nan': 'Negative', 'product': 'Negative'}
Review: expected better product suction good heavy noise
Aspects: {'product': 'Positive', 'suction': 'Positive', 'heavy': 'Positive', 'noise': 'Positive'}
Review: slightly disappointed small size
Aspects: {}
Review: could way better poor branding damage place disc brake tyre ring please friend buy product also provide cycle bell lock power break fashion cycle poor
Aspects: {'way': 'Negative', 'better': 'Negative', 'branding': 'Negative', 'damage': 'Negative', 'fashion': 'Negative', 'cycle': 'Negative'}
Review: wonderful super
Aspects: {}
Review: super nice
Aspects: {}
Review: value money using product last month delivered flipkart satisfied product easy install use problem last one month
Aspects: {'satisfied': 'Positive', 'product': 'Positive', 'last': 'Negative', 'one': 'Negative'}
Review: waste money good
Aspects: {}
Review: simply awesome nice
Aspects: {'simply': 'Positive'}
```

SECTION 2: CHATBOT ON NLP CONCEPTS AND AI APPLICATIONS

Dataset : Intent Based Coolection

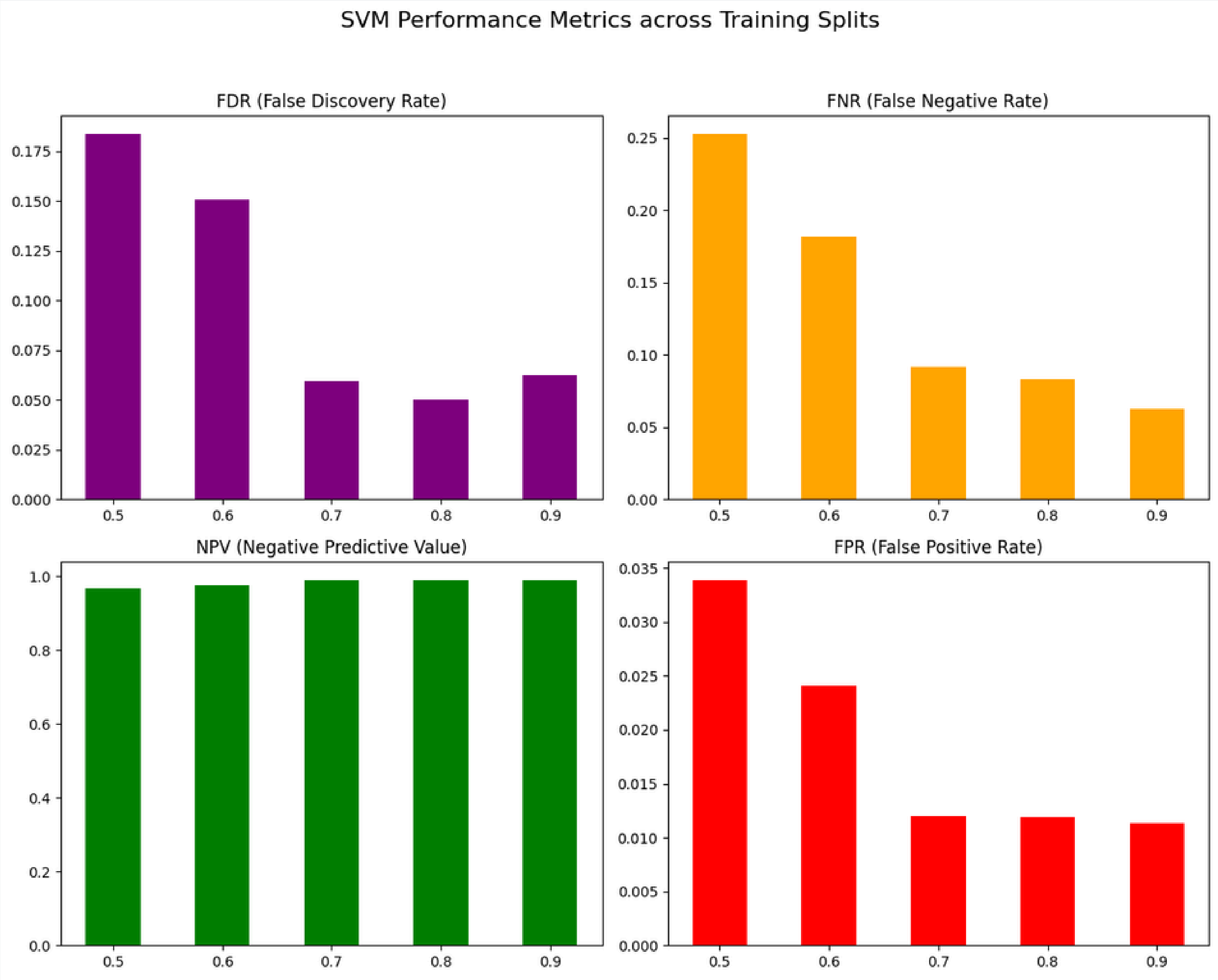
Hybrid Preprocessing:

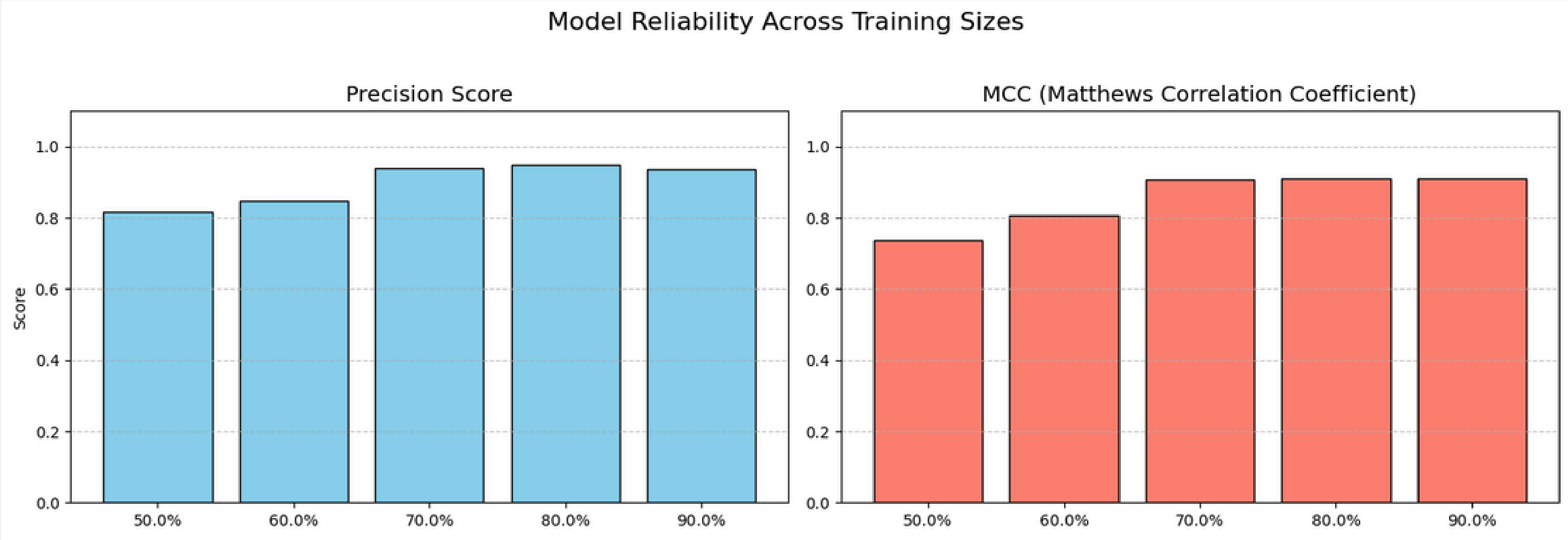
1. Get Lemmas (e.g., 'better' → 'good') via **spaCy**
2. Get Stems (e.g., 'segmentation' → 'segment') via **NLTK**

```
Data Preparation Complete!
Total Unique Patterns: 117
Labels counts: ai_application    21
tfidf_concept     20
nlp_core          18
greeting          15
goodbye           13
tokenization      11
lemmatization     10
stemming           9
Name: count, dtype: int64
```

Model	Accuracy	Strength
Linear SVM	**0.893**	**Best for clean, high-dimensional text.**
Voting Ensemble	**0.893**	Strong, but relies heavily on the SVM's input.
Logistic Regression	**0.821**	Good, but struggles with complex boundaries.
Random Forest	**0.750**	Better for tabular data than pure text vectors.
Gradient Boosting	**0.143**	**Failing** because it needs much more data to learn.

SEECTION 2: CHATBOT ON NLP CONCEPTS AND AI APPLICATIONS





HYBRID CHATBOT AND ASPECT-BASED SENTIMENT ANALYSIS ARCHITECTURE

Conversational AI and Opinion Mining for Product Reviews

```
Input: Hi how are you?
🤖 (Tag: greeting | Conf: 0.58) Hello! I'm your AI & NLP assistant.
Output: 🤖 (Tag: greeting | Conf: 0.58) Hello! I'm your AI & NLP assistant.
__________________________________________________
Input: what is NLP?
🤖 (Tag: nlp_core | Conf: 0.91) NLP (Natural Language Processing) is a field of AI that gives computers the ability to understand, interpret, and genera
Output: 🤖 (Tag: nlp_core | Conf: 0.91) NLP (Natural Language Processing) is a field of AI that gives computers the ability to understand, interpret, an
__________________________________________________
Input: what is ai?
🤖 (Tag: ai_application | Conf: 0.86) AI is used in healthcare for diagnosis, finance for fraud detection, recommendation systems like Netflix, and self
Output: 🤖 (Tag: ai_application | Conf: 0.86) AI is used in healthcare for diagnosis, finance for fraud detection, recommendation systems like Netflix,
__________________________________________________
Input: what is machine learning?
🤖 (Confidence: 0.24) I'm not 100% sure. Are you asking about NLP basics or maybe a product review?
Output: 🤖 (Confidence: 0.24) I'm not 100% sure. Are you asking about NLP basics or maybe a product review?
__________________________________________________
Input: what is deep learning?
🤖 (Confidence: 0.24) I'm not 100% sure. Are you asking about NLP basics or maybe a product review?
Output: 🤖 (Confidence: 0.24) I'm not 100% sure. Are you asking about NLP basics or maybe a product review?
__________________________________________________
Input: what is natural language processing?
🤖 (Tag: nlp_core | Conf: 0.92) NLP (Natural Language Processing) is a field of AI that gives computers the ability to understand, interpret, and genera
Output: 🤖 (Tag: nlp_core | Conf: 0.92) NLP (Natural Language Processing) is a field of AI that gives computers the ability to understand, interpret, an
```

```
__________________________________________________
Input: Camera is great but battery is bad
Output: Overall Sentiment: Positive

Aspect-Based Sentiment:
- camera: Positive
__________________________________________________
Input: Camera quality is amazing but battery drains fast
Output: Overall Sentiment: Positive

Aspect-Based Sentiment:
- camera: Positive
- quality: Positive
- fast: Negative
__________________________________________________
Input: Sound is great but build quality is poor
Output: Overall Sentiment: Negative

Aspect-Based Sentiment:
- sound: Positive
__________________________________________________
Input: Display is excellent but performance is slow
Output: Overall Sentiment: Positive

Aspect-Based Sentiment:
- display: Positive
__________________________________________________
Input: This product is amazing and works perfectly
Output: Overall Sentiment: Positive

Aspect-Based Sentiment:
- product: Positive
- work: Positive
- perfectly: Positive
__________________________________________________
Input: Excellent quality and great value for money
Output: Overall Sentiment: Positive

Aspect-Based Sentiment:
- quality: Positive
- value: Positive
- money: Positive
__________________________________________________
Input: Very bad product, stopped working in one week
Output: Overall Sentiment: Negative

Aspect-Based Sentiment:
- product: Negative
- stopped: Negative
```

THANK YOU

Anwesha!! & Oshika !!

we did a lot of efforts